

Privacy

1. Introduction

- Def. of privacy

the claim of individuals, groups and institutions to determine for themselves, when, how and to what extent information about them is communicated to others

- 3 dimensions of privacy:

- 1) Personal privacy

Protecting a person against undue interference (such as physical searches) and information that violates his/her moral sense

- 2) Territorial privacy

Protecting a physical area surrounding a person that may not be violated without the acquiescence of the person

- Safeguards: laws referring to trespassers search warrants

- 3) Informational privacy

Deals with the gathering, compilation and selective dissemination of information

Why Is Privacy Important?

- Data / Information is a valuable asset, like any other
- Data are at a higher risk of theft or misuse than ever before
- Companies have obligations to protect data
 - Laws, regulations, guidelines
 - Contracts with third parties
 - Privacy policies for users of websites, other online features

Free Email!

The Google logo, featuring the word "Google" in its signature multi-colored font.

They are happy to give email away for free, the data they collect is worth millions.

The Yahoo! logo, featuring the word "YAHOO!" in a bold, red, serif font.

Their free email comes with a price, your privacy!

The Bing logo, featuring the word "bing" in a blue, lowercase, sans-serif font.

Some even charge for advanced features, upgrade now for the privilege of paying them to invade your privacy.

Your mail and so much more

Get a whole year for \$19.99—that's less than \$2/month!

[Upgrade Now](#)



SeaNSun LLC

Small Business Solutions

www.seansunllc.com

No need for a warrant - requests

Google gives user data to government in most cases

0 Comments



NEW

Portfolio Relevance
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By John Letzing, MarketWatch

SAN FRANCISCO (MarketWatch) — Google Inc. disclosed on Monday that it complied with 94% of government requests for data about users of its services in the U.S. in the second half of last year, shedding some light on the Internet search firm's behind-the-scenes interaction with law enforcement.

Email Privacy

- Although email is a revolutionary invention, it still has some flaws.
- Most emails are not encrypted, so can be hacked
- Email has to go through another computer to reach you, so it may be intercepted and read
- Your internet service provider stores copies of emails you have received which may be read if required permission
- Headers and forwarded messages can reveal your email address to anyone that receives the message
- So, in fact, email is not very 'private' at all.

What is Web Privacy?

- Web privacy is the concept of how to maintain certain standards of control over information procured, used and stored on the Internet.
- In other words, web privacy keeps data about a person hidden when accessing websites or simply surfing the web.
- How to provide personalization of Internet services while still protecting users privacy
- → when we [read information or chat on a web site](#), how can we ensure that our privacy is preserved?
- → when we [search for information with a search engine](#), how can we ensure that traces of our searches will not be used for purposes outside our control?
- → when we [register on a website](#), how can we ensure that private data will not be disseminated on the web to other parties we do not trust?

What's their profit???

- Why are the companies worth so much money?
- Why is MySpace, Facebook, Flickr worth over half a billion dollars without a proper revenue model?

How 'they' can infiltrate our privacy?

- Information we gather and track
- Personal information
- Aggregated Tracking Information
- Cookies
- Third party cookies
- Third-party advertising

3. Threats to Privacy

- 1) Threats to privacy at application level
- 2) Threats to privacy at communication level
- 3) Threats to privacy at system level
- 4) Threats to privacy in audit trails

❖ Examples

- Unauthorized use by “insiders”
- Inadvertent leakage through negligence, carelessness, hacking
- Propagation of errors
- Intentional: marketing, decision making, surveillance
- Cross-matching files

Information: Government and private databases

Individually, most do not pose much threat to privacy but Collectively, a lot!

- Ministry data
- National Security
- Pension Funds
- Income Tax
- Banks
- Hospitals and Sick Funds
- Schools & universities
- Libraries
- Club memberships
- Employers
- Airlines and travel agencies
- Car registration
- Telephone companies
- Census

Database Nation

- Tiny chunk of data can reveal your home address
- Linking to People searches can reveal your phone number
- Your county government's Web displays your home's floor plan and assessed value
- Google and Yahoo! can record every search you've ever made and link it to whatever computer you used at the time.
- Credit card companies know what you buy
- frequent shopper programs know what you eat.
- Your insurance company knows what medical procedures you've undergone.
- In a February 2003 Harris poll, 69 percent of those surveyed agreed that "consumers have lost all control over how personal information is collected and used by companies."
- Sun Microsystems CEO Scott McNealy. "You have zero privacy anyway," he said a few years ago.

Hacking

- ❖ Hacking tips are widely available
- ❖ Threat to privacy
- ❖ Trying different passwords is trivial hacking
 - *use hard-to-crack password!*

Monitoring

- ❖ *Employee Monitoring* -
how much should one tolerate?

Consumer Information

- ❖ Released through:
 - Marketing survey
 - Making donations
 - Making purchases
 - “Transaction-generated information”
- ❖ Data mining to find out more
- ❖ Often collected without awareness
 - *Invisible information gathering*
 - cookies
- ❖ Result may cause inconvenience, embarrassment and threat
 - e.g. list of consumers who are likely to buy products for adults who are incontinent was made public
 - junk mail

Medical Records

- ❖ How much is protected?
 - ❖ Can insurance company access everything?
 - Can one check on AIDS patients?

Wiretapping

- ❖ Interception of a communication in a telephone or computer network
- ❖ Telephone bugging now extends to wiretapping on fax transmission, email, voice mail and computer networks
- ❖ Account numbers, passwords, PINs used to be obtainable from touch tones; now they are in emails, or web communications

Privacy Protection

- ❖ Technical : Encryption, Digital Signature, Watermarking, steganography
- ❖ Operational: Access Control, empty cache and cookies
- ❖ Behavioral: Wiping data after completion of task, Setting strong password and keep it changing, Search with a proxy, Avoid registrations, Disable auto-complete, be Careful while downloading
- ❖ User Identity
- ❖ Law

Encryption

Meaning:

- ❖ Cryptography – “the art and science of hiding data in plain sight”
- ❖ To transform a message or data into a form that is meaningless to anyone who might intercept it.

Uses:

- Electronic transfer of funds
- Passwords and PINs for consumer transaction
- Credit card numbers
- Bank records and financial data
- Sensitive business communications
- Research and product information
- Personal data and communications
- Test results

Encryption

- ❖ Cyphertext – the coded message
- ❖ Decryption – the decoding process
- ❖ Key – a specific sequence of characters for coding/decoding

Public Key Cryptography

- ❖ Beforehand, key must be known to both parties
- ❖ Public key and private key
 - Encrypting key is made public
 - Public key (of the recipient) is used to encrypt message sent
 - Private key is used to decrypt by the recipient

PGP (Pretty Good Privacy)

- ❖ A program developed by Philip Zimmermann using public key cryptography for emails
- ❖ Provides protection of emails from wiretapping
- ❖ Widely and freely available

Digital Signatures

- ❖ Some public key cryptography schemes allow the reversal of public and private keys to provide digital signatures
 - A person accepts an electronic document and encrypts it with his private key. Recipient can then decrypt the document with the public key.

Protecting User Identity

- **Anonymity** - a user may use a resource or service without disclosing her identity
- **Pseudonymity** - a user acting under a pseudonym may use a resource or service without disclosing his identity
- **Unobservability** - a user may use a resource or service without others being able to observe that the resource or service is being used
- **Unlinkability** - sender and recipient cannot be identified as communicating with each other

Anonymity and Pseudonymity

- ❖ Hide identity for self-protection:
 - voice out against government
 - voice out against employer
 - “off” the record
- ❖ Use pseudonymous remailers
- ❖ Programs available for anonymous Web browsing, email encryption, and chat room anonymity

Privacy Policy-1

1. There must be a way for an individual to **find out what information** about him or her is in a record and how it is used.
2. There must be a **way for an individual to correct** or amend a record with information that is identifiable to him or her.
3. The existence of all data systems with personal information in them should be **publicly disclosed**, and the purpose for which information is gathered about people should be disclosed.
4. There must be a **way for an individual to prevent information** about him or her that was obtained for one purpose (which was stated when the information was gathered) from being used or made available, either within the organization or outside, for a purpose that is incompatible with the original purpose, without getting the consent of the individual. This is the principle of secondary use.

Privacy Policy-2

5. The organization creating, maintaining, using, or disseminating records of identifiable personal data **must assure the reliability, accuracy, security and timeliness of the data**. In other words, the custodian of information that is disseminated has an obligation to the individual to make sure it is accurate, secure, and not misused. This obligation ought not be delegated to another entity
6. An **organization must conduct periodic risk assessments, balancing the possibility** or probability of unauthorized access or disclosure against the cost of security precautions and the expected effectiveness of the precautions.
7. An organization must make sure that other entities handling personal information in behalf of the first **organization are bound by these same principles**.

Privacy Policy-3

8. Organizations must take **special precautions in collecting and using personal information** about children for those 18 and younger.
9. An organization should **openly disclose its policies and practices with regard to electronic surveillance** of its employees' and customers' telephone calls, electronic mail, Internet usage, changing rooms, and rest rooms, and must articulate in advance the reasons for the surveillance.

Privacy Protection Under the Law

While currently there is no national law to protect the privacy of the information you share online, federal law and state law do offer some protection to various kinds of personal information collected about you.

Offensive Speech and Censorship in Cyberspace

Speech Might Include:

- Political or religious speech.
- Pornography (Multibillion-dollar online business).
- Sexual or racial slurs.
- Nazi materials.
- Libelous statements.
- Abortion information.
- Alcohol ads.

Material is considered obscene if all three parts are met:

1. It depicts sexual (or excretory) acts whose depiction is specifically prohibited by state law, and
2. It depicts these acts in a patently offensive manner, appealing to the prurient interest as judged by a reasonable person using community standards, and
3. It has no serious literary, artistic, social, political, or scientific value.

Offensive Speech and Censorship in Cyberspace

Material Inappropriate for Children

Technology Changes the Context

- On the Web, children have access to the same 'adult' text, images, videos, etc. as adults (no "ticket taker" to check age).
- Online proprietors don't know the customer is not an adult.

Protecting Children

Regardless of the medium:

- It is illegal to create, possess or distribute child pornography. (Because creation of it is child abuse...)
- It is illegal to lure children into sexual activity.

Protect Children

Software filtering

Bess and Cyber Patrol

Offensive Speech and Censorship in Cyberspace

Limiting Internet Access in Libraries and Schools

Filtering Software

- Benefit: prevent access to inappropriate material on the Internet by screening words or phrases, blocking sites according to rating system, or disallowing access to specific sites in a list.
- Problems: can be ineffective—kids get around the filters; the words, phrases, rating systems, etc. are subjective; “banned” keywords can be overly restrictive for adult users and for legitimate use by minors.

Challenging Old Regulatory Paradigms and Special Interests

License required:

- To practice law.
- To publish traditional newsletters about commodities and futures investing (prior to 2000).

No license required:

- To create downloadable, self-help legal software.
- To publish newsletters about, developing software for, and operating Web sites concerning commodities and futures investing (since 2000).